Course Outline

CSIT 175 - Python Programming

Fall Semester 2026

Basic Course Information

Courses numbered 1 - 49 are remedial or college preparatory courses which do not apply toward an A. A. Degree and are not intended for transfer. Courses numbered 50-99 apply toward an AA Degree, but are not intended for transfer. Courses numbered 100 and higher apply toward an AA Degree and/or are intended for transfer to a four-year college or university.

CB04 Course Credit Status D - Credit - Degree Applicable

Discipline CSIT

Title Python Programming

This course is part of the following cross listing

Units and Hours

Min Unit

3

Grade Option Grade/Pass/No Pass

Min Semester Hours Calculated

Hour Type	Units	Weekly Hours	Semester Hours x 16 Weeks	Semester Hours x 18 Weeks
Lecture Category -	2.00	2.00	x 16 Weeks - 32.00	x 18 Weeks - 36.00
Lab Category -	1.00	3.00	x 16 Weeks - 48.00	x 18 Weeks - 54.00
Subtotal -		5.00	x 16 Weeks - 80.00	x 18 Weeks - 90.00
Out of Class Hour -		4.00	x 16 Weeks - 64.00	x 18 Weeks - 72.00
Totals -		9.00	x 16 Weeks - 144.00	x 18 Weeks - 162.00

Basic Skills Requirements: Appropriate Language and/or Computational Skills.

Catalog Description

Students will gain the fundamental knowledge and skills necessary to use the Python programming language to develop software for applications. Topics of study include the fundamentals of the language, using lists, dictionaries and creating functions, Object-Oriented programming and the use of AI to code, enhance and troubleshoot programs.

Student Learning Outcomes

Outcome

1. Design and develop Python programs that meet specified requirements.
2. Demonstrate professional Python code formatting practices that allow code to be easily maintained and understood. The practices are spelled out in the PEP 8 Style Guide for Python Code.

Specific Course Objectives

Objective

Upon successful completion of the course, the student will be able to:

- Understand the fundamentals of the Python programming language;
- Write syntactically and semantically correct Python programs;
- Analyze program requirements and design solutions using Python;
- Use an AI Chatbot to code, enhance and troubleshoot programs.

Methods of Instruction

Methods of Instruction may include, but are not limited to, the following:

- Discussion
- Lab
- Lecture

Content in Terms of Specific Body of Knowledge

I. Variables and Data Types
   A. Naming and Using Variables
   B. Changing Case with Strings
   C. Concatenation
   D. Number Handling
II. Working with Lists
   A. Changing, Adding, and Removing Elements
   B. Organizing a List
   C. Looping Through a List
   D. Tuples
III. Decision Statements
   A. Conditional Tests
   B. IF statements
   C. Using IF statements with Lists
IV. Dictionaries
   A. Accessing Values in a Dictionary
   B. Looping through Dictionary
V. User Input and Loops
   A. Using the input() Function
   B. While loops
   C. Using a Loop with Lists and Dictionaries
VI. Functions
   A. Defining a Function
   B. Passing Arguments

C. Return Values
D. Passing a List
E. Storing Functions in Modules

VII. Object-Oriented Programming
A. Creating and Using a Class
B. Classes and Instances
C. Inheritance
D. Importing Classes

VIII. Files and Exceptions
A. Reading from a File
B. Writing to a File
C. Exceptions
D. Storing Data

IX. Using an AI Chatbot
A. Enhance Python program using an AI Chatbot
B. Troubleshoot a non-working Python program using an AI Chatbot
C. Develop a new Python program using an AI Chatbot

Textbooks/Resources

Textbook

1. **Author** Perry, Stephen
   Title Painless Python
   Edition 1st
   City San Diego
   Publisher Cognella Academic Publishing
   Year 2023

Assignments

Students are expected to research projects and solutions as dictated by specific assignments. Students are expected to spend a minimum of three hours per unit per week in class and on outside assignments, prorated for short-term classes.

Required Reading
Students will be required to read the textual material provided on the class web site and/or from the required book.

Suggested Reading

Required Writing
Students will explain what troubleshooting approaches they found must useful and share those findings with the other students in the class.

Critical Thinking
All of the courses in computer programming require students to learn to think critically. Using computers effectively requires that students can express their instructions in a form that the computer program can understand and execute. First students must understand what they want to accomplish. Second, students

must identify what logical steps are required to instruct the computer to do the work. Finally, students must know how to submit the instructions in the correct format to the computer for completion. Accomplishing this work requires students apply critical thinking skills.

Outside Assignments
Outside assignments will include completion of lab work, assigned readings the text books, and homework problems.

Methods of Assessment

Evaluation Method

- Class Participation
- Exams/Tests
- Homework
- Projects
- Quizzes

Open Entry/Open Exit

Open Entry / Open Exit - Not Open Entry/Open Exit

Repeatability

Is this course repeatable in accordance with Title 5 guidelines? No

Contact Person

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